Ayush Kumar

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Summary

Final-Year B.Tech Information Technology student at IIIT Bhubaneswar specializing in Data Structures & Algorithms, Full-Stack MERN Engineering, and Applied AI/ML. Experienced in architecting scalable RESTful APIs, autonomous agent workflows (LangGraph), vector search pipelines (pgvector), and reliable CI/CD microservices with Docker and GitHub Actions, targeting Software Development Engineer (SDE) roles.

Education

International Institute of Information Technology (IIIT) — **Bhubaneswar, Odisha**
Bachelor of Technology in Information Technology (CGPA: 7.48) — *2023 – 2027*

Delhi Public School, Nalco Nagar — **Angul, Odisha**
Class XII (Senior Secondary): 88.6% | Class X (Secondary): 98.2% — *2020 – 2023*

Technical Skills

Languages: Java, JavaScript (ES6+), Python, TypeScript, SQL, HTML5, CSS3
Web Technologies: React.js, Node.js, Express.js, FastAPI, Flask, Vite, Tailwind CSS, Bootstrap 5, EJS, RESTful APIs
AI & Machine Learning: LangGraph, RAG Pipelines, pgvector, Scikit-learn, XGBoost, Pandas, NumPy, Google Gemini API
Databases & DevOps: PostgreSQL, MongoDB Atlas, Docker, Docker Compose, Git, GitHub Actions (CI/CD), Cloud Computing (Basics), Render, Cloudinary
Core CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming (OOP), DBMS, Operating Systems

Projects

Feasto | *React.js, Node.js, Express.js, MongoDB Atlas, Razorpay, Docker, CI/CD* — GitHub | Live Demo

- Engineered a full-stack MERN food delivery app with decoupled customer and admin portals, managing persistent cart state and real-time GPS delivery tracking via Leaflet.
- Integrated Razorpay payments with HMAC-SHA256 webhook signature verification, delivering idempotent checkout processing with zero payment discrepancies, and secure JWT role-based access control (RBAC).
- Embedded an AI food concierge using the Google Gemini API for tailored meal recommendations; provisioned multi-stage Docker builds and continuous GitHub Actions CI/CD for cloud deployment on Render.

StayNest | *Node.js, Express.js, MongoDB Atlas, EJS, Bootstrap 5, Cloudinary, Razorpay, PDFKit* — GitHub | Live Demo

- Developed an MVC-architecture accommodation marketplace with full CRUD listing management, geospatial coordinate filtering, customer review moderation, and responsive Bootstrap 5 UI.
- Designed a collision-free booking calendar with blackout scheduling, authenticated Razorpay payment capture, and programmatic server-side invoice generation via PDFKit with Joi input validation schemas.
- Hardened security with Passport.js session auth and Helmet HTTP headers; configured MongoDB Atlas persistence with Cloudinary CDN and continuous GitHub Actions CI/CD regression testing.

Smart House Price Predictor | *Python, Flask, Scikit-learn, XGBoost, Docker, CI/CD* — GitHub | Live Demo

- Trained an end-to-end real estate valuation model on 20,000+ property records, achieving $R^2 = 83.93\%$ and 17.52% MAPE ($30,245 MAE) with an optimized XGBoost gradient-boosted ensemble.
- Formulated predictive confidence intervals and feature importance attribution across 13 demographic and geographic housing variables for transparent valuation insights.
- Delivered real-time inferences through a low-latency (<20ms) Flask RESTful API with defensive Pydantic validation, packaged in Docker with scheduled GitHub Actions CI/CD test runs.

AgentIQ | *Python, FastAPI, LangGraph, PostgreSQL, pgvector, React, Docker, CI/CD* — GitHub | Live Demo

- Architected an autonomous agentic RAG platform using FastAPI and LangGraph, orchestrating bounded multi-step reasoning workflows (≤ 4 steps) with deterministic fallbacks and sub-second retrieval latency.
- Built hybrid semantic search on PostgreSQL 16 and pgvector with HNSW indexing and cosine similarity cutoffs, returning source citations for every generated answer, protected by parameterized SQL validation and RBAC.
- Constructed a responsive 3-pane dashboard in React 18 and TypeScript; orchestrated multi-tier microservices with Docker Compose and established continuous GitHub Actions CI/CD test pipelines.

Achievements & Certifications

- Completed **Alpha (Data Structures & Algorithms with Java)** certification by Apna College (View Certificate); mastered core Data Structures, Algorithms, Dynamic Programming, and OOP principles.
- Solved **200+ Data Structures & Algorithms** problems in Java across **LeetCode** and **GeeksforGeeks**, demonstrating strong algorithmic proficiency in graph algorithms, dynamic programming, and complexity analysis.
- Shipped and deployed **4 production-grade systems** (Feasto, StayNest, SmartHouse, AgentIQ) leveraging **Docker**, **GitHub Actions CI/CD pipelines**, and **cloud hosting on Render**.
- Demonstrated sustained academic excellence with **98.2% in Class X** and **88.6% in Class XII**; actively pursuing B.Tech in Information Technology at IIIT Bhubaneswar.